

BORGWARNER

CM Inverter Software Errata

Document: 0A-0174-01

BorgWarner Portland (formerly Cascadia Motion) CM Inverter Software Errata

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1 Summary

This document captures deviations from design intent for the CM series inverter software. Each bug will include a detailed description of how the bug manifests itself, what it affects, any known workarounds and the solution to fixing it. Each bug is also accompanied by a user impact assesment.

1.1 Software Versions

Table 1: Inverter/Software Versions

Software Version	Release Date
v6520	May 13th, 2022
v6521	May 16th, 2022
v6522	July 17th, 2022
v6523	July 21st, 2022
v6524	September 13th, 2022
v6525	September 27th, 2022
v6526	December 1st, 2022
v6527	February 28th, 2023
v6528	April 11th, 2023
v6529	June 6th, 2023
v652A	June 14th, 2023
v652B	August 23rd, 2023
v652C	August 29th, 2023
v652D	November 17th, 2023
v652E	March 21st, 2024
v652F	April 3rd, 2024
v6530	May 24th, 2024
v6531	August 29th, 2024
v6532	February 6th, 2025
v6533	May 14th, 2025

1.2 User Impact Assesment

Table 2: User Impact Assesment Level

Impact Level	Description
High	The user needs to update their configuration settings or will be affected by this change.
Medium	The user may need to update their configuration settings if using this feature or is likely to be affected by this change.
Low	The user does not need to update their configuration settings or is unlikely to be affected by this change.

2 Fixes in v6533

2.1 Inverter State Machine: Oscillation between states 9 and 1

Impact Level	Versions Affected
Low	v6520-v6532

Description:

The user may see an oscillation between states 9 and 1 of the inverter state machine variable made available over the CAN signal 'INV_Inverter_State' in the 'M170_Internal_States' message during fault conditions.

Solution Implemented:

A lockout mechanism has been added on the transition from states 9 to 1, only allowing the transition to happen once. This lockout is cleared upon the transition of state 9 to states 2 or 3.

2.2 Inverter State Machine: Torque Command Persists in Stop State

Impact Level	Versions Affected
Low	v6520-v6532

Description:

The inverter torque command set by the torque rate limiters persists through enable/disable cycles resulting in incorrect torque output when using low torque rate limits.

Solution Implemented:

The inverter torque command is cleared upon the entrance of inverter state 0.

2.3 SCI Command: EXC_CMD Not changing

Impact Level	Versions Affected
Low	v6520-v6532

Description:

In some instances, sending a value to the defsyms command 'Exc_Filter_Command_(BITS_2_1_0)' will appear to be accepted when it is not. This might affect users who are configuring their inverter to work with motor type 65, during resolver tuning.

Solution Implemented:

Inverters that do not support dynamically setting the resolver excitation level will not allow this value to be changed via serial. If a user is performing resolver calibration for motor type 65 they should rely on using the defsyms 'Resolver_Exc_Level_EEPROM'.

2.4 Variable PWM Setting: EEPROM Write Via CAN Corrupting EEPROM

Impact Level	Versions Affected
Low	v6526-v6532

Description:

If an invalid PWM setting is written over CAN, the CAN EEPROM interface attempts to correct it. In certain instances, this can lead to two parameters being updated, but storing only one at a time via CAN EEPROM writing, causing a checksum error.

Solution Implemented:

If an invalid PWM setting is written over CAN, the setting is discarded, and the last known valid settings are used.

2.5 Variable PWM Setting: EEPROM Write Via CAN Appears Valid

Impact Level	Versions Affected
Low	v6526-v6532

Description:

Writing variable PWM settings via CAN always returns a successful response, even if the setting is invalid.

Solution Implemented:

Invalid settings are discarded and the CAN Parameter Write message returns a failed status.

2.6 Torque Ramp Rate: Torque Limits Violated on Direction Switch

Impact Level	Versions Affected
Low	v6520-v6532

Description:

The torque rate limiter module do not account for direction deviating from commanded direction when applying torque limits. This is particularly apparent on systems using low torque ramp rates and only allowing for either motoring or regeneration torque.

Solution Implemented:

The torque rate limiter module takes into account direction when applying torque limits. Additionally, torque command is now reset during a brief disable event, required when switching directions.

2.7 Torque Ramp Rate: Torque Command Ramp Rate

Impact Level	Versions Affected
Low	v6520-v6532

Description:

The torque rate limiter module doesn't clear existing torque command output on a disable event which can cause a torque command to be present when re-enabling. This is particularly apparent on systems using low torque ramp rates with torque commands such that the torque rate limiter module fails to adequately manage the transition between enabled and disabled states.

Solution Implemented:

The torque rate limiter module overwrites torque rate when disabling such that zero torque will be reached within the disable time. The user's torque rates will be restored when re-enabling.

2.8 Shudder Compensation: Regen Torque Limit Violation

Impact Level	Versions Affected
Low	v6520-v6532

Manifestation:

Shudder compensation modes 1 and 2 can exceed the limit set by the torque capability feature, allowing an excess regen torque to reach the maximum regen torque limit set in EEPROM. This excess torque can occur if shudder compensation is active near the current regen torque capability.

Solution Implemented:

Shudder compensation correctly obeys limitations set by torque capability feature, allowing for torque overhead to be applied.

2.9 Torque EEPROM Limits: Limit swap on reverse direction

Impact Level	Versions Affected
Low	v6532

Description:

The motoring and regen torque limits configured in an inverter's EEPROM settings will be swapped during a reverse direction. This only applies to motors using lookup table based Id/Iq currents.

Solution Implemented:

The Id/Iq current table lookup is now bounded by the correct torque limitations.

3 Revision History

Version	Description	Updated By	Date
0A-0174-01	Initial Content, v6533. ECO 1142.	Noah Erickson	05/14/2025